UNDER THE ZARISKIAN SECTION CONDITIONS OF GABBER, SECTIONS OF AN ETALE SHEAF OVER X AGREE WITH SECTIONS OVER THE CLOSED SUBSCHEME Z

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full, with the injectivity computation, the local description of an arbitrary section t, the finite cover X' -> X, the lift by assumption (2) and the closing finite-descent step. The one passage left to the reader is the routine compatibility of the construction i^{-1}s with pullback along morphisms of X_etale and along finite X' -> X, which the delivered text states and which plays no role in the argument that follows.

Editorial note. The label lemma-gabber-h0 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The statement is the author's, copied from the pinned chapter file; the conditions (1) and (2) are part of the statement and are delivered verbatim.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Small etale site pullback i^{-1}_{small} along a closed immersion. Same chapter, the pullback i^{-1}_{small} of an etale sheaf along a closed immersion i : Z -> X and its description on stalks. Supplies the object whose sections over Z are compared with the sections of F over X.
- Zariski and etale pullbacks of a sheaf. Same chapter, the restriction F|_{X_{Zar}} to the Zariski site of X, the pullback i^{-1} on Zariski sheaves, and the base change map between them. Supplies the canonical comparison map that the proof shows to be injective on stalks and then identifies with a subsheaf.
- Elementary etale neighbourhoods and etale coverings. Same chapter, elementary etale neighbourhoods (U, u) -> (X, z) and the sheaf condition for the etale topology. Supplies the colimit description of the stalk of i^{-1}_{small}F and the injectivity of F(U_0) -> F(U).

Source reference. Original lemma 09Z0: sources/stacks-chow/tags.txt, source lines 12583–12583. Prerequisite pointer invoked at proof line(s) 15038; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 09Z3: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8488–8488. Prerequisite pointer invoked at proof

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line(s) 15068; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00VD: sources/stacks-chow/tags.txt, source lines 1143–1143. Prerequisite pointer invoked at proof line(s) 15018; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme. Let $i : Z \to X$ be a closed immersion. Assume that*

(1) *for any sheaf $\mathcal{F}$ on X_{Zar} the map $\Gamma(X, \mathcal{F}) \to \Gamma(Z, i^{-1}\mathcal{F})$ is bijective, and*
(2) *for any finite morphism $X' \to X$ assumption (1) holds for $Z \times_X X' \to X'$.*

Then for any sheaf $\mathcal{F}$ on $X_{étale}$ we have $\Gamma(X, \mathcal{F}) = \Gamma(Z, i^{-1}_{small}\mathcal{F})$.

Proof. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. There is a canonical (base change) map

$$i^{-1}(\mathcal{F}|_{X_{Zar}}) \longrightarrow (i^{-1}_{small}\mathcal{F})|_{Z_{Zar}}$$

of sheaves on Z_{Zar}. We will show this map is injective by looking at stalks. The stalk on the left hand side at $z \in Z$ is the stalk of $\mathcal{F}|_{X_{Zar}}$ at z. The stalk on the right hand side is the colimit over all elementary étale neighbourhoods $(U, u) \to (X, z)$ such that $U \times_X Z \to Z$ has a section over a neighbourhood of z. As étale morphisms are open, the image of $U \to X$ is an open neighbourhood U_0 of z in X. The map $\mathcal{F}(U_0) \to \mathcal{F}(U)$ is injective by the sheaf condition for $\mathcal{F}$ with respect to the étale covering $U \to U_0$. Taking the colimit over all U and U_0 we obtain injectivity on stalks.

It follows from this and assumption (1) that the map $\Gamma(X, \mathcal{F}) \to \Gamma(Z, i^{-1}_{small}\mathcal{F})$ is injective. By (2) the same thing is true on all X' finite over X.

Before we prove the surjectivity, let us introduce some notation. For any object U of $X_{étale}$ and $s \in \mathcal{F}(U)$ we denote $i^{-1}s \in (i^{-1}_{small}\mathcal{F})(U \times_X Z)$ the pullback (more precisely this is the image of s under the map of Sites, Lemma 00VD combined with the map from the presheaf pullback to the sheaf pullback). This construction is compatible with pullback by morphisms in $X_{étale}$ and by (finite) morphisms $X' \to X$ in a manner which we leave to the reader to clarify.

Let $t \in \Gamma(Z, i^{-1}_{small}\mathcal{F})$. By construction of $i^{-1}_{small}\mathcal{F}$ there exists an étale covering $\{V_j \to Z\}_{j \in J}$, étale morphisms $U_j \to X$, sections $s_j \in \mathcal{F}(U_j)$ and morphisms $V_j \to U_j$ over X such that $t|_{V_j}$ is the pullback of $i^{-1}s_j$ by $V_j \to U_j \times_X Z$. Since $V_j \to U_j \times_X Z$ is étale, the image is an open subscheme. Since $U_j \times_X Z \subset U_j$ is closed, we may, after replacing U_j by an open subscheme, assume that $V_j \to U_j \times_X Z$ is surjective. Then $\{V_j \to U_j \times_X Z\}$ is an étale covering and we conclude that $t|_{U_j \times_X Z} = i^{-1}s_j$. Observe that every nonempty closed subscheme $T \subset X$ meets Z by assumption (1) applied to the sheaf $(T \to X)_*\underline{\mathbf{Z}}$ for example. Thus we see that $\coprod U_j \to X$ is surjective. By More on Morphisms, Lemma 09Z0 we can find a finite surjective morphism $X' \to X$ such that $X' \to X$ Zariski locally factors through $\coprod U_j \to X$. Say $X' = \bigcup_{\alpha \in A} W_\alpha$ is an open covering and $j : A \to J$ is a map, such that there exist morphisms $g_\alpha : W_\alpha \to U_{j(\alpha)}$ over X. Denote $\mathcal{F}'$ the pullback of $\mathcal{F}$ to $X'_{étale}$. Denote $s'_\alpha \in \mathcal{F}'(W_\alpha)$ the pullback of $s_{j(\alpha)}$ by g_α. Set $Z' = X' \times_X Z$. Denote $i' : Z' \to X'$ the base change of i. Then $(i'_{small})^{-1}\mathcal{F}' = (Z' \to Z)^{-1}_{small} i^{-1}_{small}\mathcal{F}$. Via this identification we may denote $t' \in \Gamma(Z', (i')^{-1}_{small}\mathcal{F}')$ the pullback of t to Z'. Then $t'|_{W_\alpha \times_X Z}$ is equal to $(i')^{-1}s'_\alpha$ by construction. We conclude that t' is a section of the subsheaf

$$(i')^{-1}(\mathcal{F}'|_{X'_{Zar}}) \subset (i'_{small})^{-1}\mathcal{F}')|_{Z'_{Zar}}$$

By assumption (2) the section t' comes from a section $s' \in \Gamma(X', \mathcal{F}'|_{X'_{Zar}}) = \Gamma(X', \mathcal{F}')$. By the injectivity proved in the second paragraph, we conclude that the two pullbacks of s' to $X' \times_X X'$ are the same (after all this is true for the pullbacks of t' to $Z' \times_Z Z'$). Hence we conclude s' comes from a section of $\mathcal{F}$ over X by Remark 09Z3. $\square$